

43. Following are spectroscopic and other data for several compounds. Propose a structure for each of them. **(a)** Molecular formula = $C_6H_4Br_2$. 1H NMR spectrum A. ^{13}C NMR: 3 peaks; IR: $\tilde{\nu} = 745$ (s, broad) cm^{-1} ; UV: $\lambda_{max}(\epsilon) = 263(150)$, 270(250), 278(180) nm. **(b)** Molecular formula = C_7H_7BrO . 1H NMR spectrum B. ^{13}C NMR: 7 peaks; IR: $\tilde{\nu} = 680$ (s) and 765 (s) cm^{-1} . **(c)** Molecular formula = $C_9H_{11}Br$. 1H NMR spectrum C. ^{13}C NMR (DEPT): $\delta = 20.6(CH_3)$, $23.6(CH_3)$, $124.2(C_{quat})$, $129.0(CH)$, $129.0(CH)$, $136.0(C_{quat})$, $137.7(C_{quat})$ ppm.

29. The following spectroscopic data are for two carbonyl compounds with the formula $C_8H_{12}O$. Suggest a structure for each. The letter "m" stands for the appearance of this particular part of the spectrum as an uninterpretable multiplet. **(a)** 1H NMR: $\delta = 1.60$ (m, 4 H), 1H NMR : $\delta = 1.60$ (m, 4 H), 2.15 (s, 3 H), 2.15 (s, 3 H), 2.19 (m, 4 H), 2.19 (m, 4 H), 6.78 (t, 1 H) ppm; ^{13}C NMR (DEPT) : $\delta = 21.6$ (CH₂), 22.0 (CH₂), 22.0 (CH₂), 23.0 (CH₂), 23.0 (CH₂), 25.1 (CH₃), 25.1 (CH₃), 26.1 (CH₂), 26.1 (CH₂), 139.7 (C_{quat}), 139.7 (C_{quat}), 140.9 (CH), 140.9 (CH), 199.2 (C_{quat}) ppm. **(b)** 1H NMR: $\delta = 0.94$ (t, 3 H), 1H NMR : $\delta = 0.94$ (t, 3 H), 1.48 (sex, 2 H), 1.48 (sex, 2 H), 2.21 (q, 2 H), 2.21 (q, 2 H), 5.8–7.1 (m, 4 H), 5.8–7.1 (m, 4 H), 9.56 (d, 1 H) ppm; ^{13}C NMR (DEPT) : $\delta = 13.6$ (CH₃), 21.9 (CH₂), 21.9 (CH₂), 35.2 (CH₂), 35.2 (CH₂), 129.0 (CH), 129.0 (CH), 135.2 (CH), 135.2 (CH), 146.7 (CH), 146.7 (CH), 152.5 (CH), 152.5 (CH), 193.2 (CH) ppm.

31. Following are spectroscopic and analytical characteristics for an unknown substance. Propose a structure. Empirical formula: $C_8H_{16}O$. 1H NMR: $\delta = 0.90$ (t, 3 H), 1.0–1.6 (m, 8 H), 2.05 (s, 3 H), 2.25 (t, 2 H) ppm; ^{13}C NMR seven signals between $\delta = 14.0$ and 43.8 ppm, and one at 208.9 ppm; IR: 1715 cm^{-1} . UV: $\lambda_{max}(\epsilon) = 280(15)$ nm; MS: $m/z = 128$ (M^+); intensity of ($M+1$) $^+$ peak is 9% of M^+ peak; important fragments are at $m/z = 113$ ($M-15$) $^+$, $m/z = 85$ ($M-43$) $^+$, $m/z = 71$ ($M-57$) $^+$, $m/z = 58$ ($M-70$) $^+$ (the second largest peak), and $m/z = 43$ ($M-85$) $^+$ (the base peak).